**CS 6603: AI, Ethics, and Society**
**Homework Project #1: Data Profile Assignment**

Readings:
- Intro & Chapter 1: Weapons of Math Destruction (What is a Model?)
- M. Kosinski, D. Stillwell, T. Graepel, "Private traits and attributes are predictable from digital records of human behavior," Proceedings of the National Academy of Sciences Apr 2013, 110 (15) 5802-5805; DOI:10.1073/pnas.1218772110, http://www.pnas.org/content/110/15/5802

In this assignment, you'll research and explore how platforms collect and use personal data to build a profile of their users. By examining your own data profile on a platform of your choice, you'll gain insight into how companies track behavior, make assumptions and use that information for their own purposes (such as advertising, recommendations or product development). You may choose from:
- Social media platform (ex. Facebook, Tik/Tok, Instagram, Reddit)
- Commerce site (ex. Amazon, eBay)
- Streaming platforms (ex. Spotify, Netflix, YouTube)
- Tech platforms (ex. Google, Apple, Microsoft)
- Other platforms you use regularly that collect and apply data about you

Important: You are not required to use a personal social media account for this assignment. Social media platforms are provided only as examples. Any platform that collects and profiles user data may be used, including dummy or test accounts.

Your task is to review the data available about you on one of these platforms and reflect on how it shapes the "digital profile" the company has created about you.

**Privacy Note:** You will **not** be asked to submit raw datasets, account credentials, or sensitive personal information as part of this assignment. You may filter, summarize, or analyze only the subset of data you are comfortable working with. Dummy or test accounts are also acceptable.

**Step 1: Research platform and identify data**
Research a platform of your choice by investigating its privacy/data settings. Identify the types of data the platform collected. As you look through the dataset, find a collection of data that the platform has generated based on your interaction with the platform. You should be able to categorize and analyze this data for how relevant it is to your profile and interests. For example, this could be:
- A list of your interests from Google Discover (powered by Google Search)
- A list of advertisers using your activity from Instagram
- A list of recommended shows from Netflix

If you prefer not to use your personal data, you may create a dummy account or use a fictional persona to generate data intentionally. In that case, your analysis should reflect how the platform inferred interests based on the persona's behavior.

**Data Requirements:**
- The dataset size should be around 100 – 400 items.
  - If your dataset is larger than this, please select a portion of it to create a reduced dataset. The reduced dataset should be what you use for the rest of the assignment.
- The data should be "organizable" into at least 5 categories.
  - You decide on the categories based on the dataset you have selected.
  - Do not create more than 10 categories.

- You may create an "Other" category for miscellaneous items.
- Examples:
  - For a list of interests: Clothes, Food, Music, Tech, Travel
  - For a list of advertisers: Restaurants, Car Companies, Education Programs, Credit Card Companies
- The data should be sorted into 3 buckets based on how relevant it is to your interests.
  - You can name these buckets however you like, but they should show a clear hierarchy of relevancy.
  - Examples:
    - For advertisers: Relevant, Not Relevant, Way Off
    - For interests: Love, Indifferent, Hate

**Example resources of how to find your data**:
- Facebook: https://www.facebook.com/help/1701730696756992
- Instagram: https://help.instagram.com/181231772500920
- Google: https://takeout.google.com/

**Tip/Notes**:
- Some platforms allow users to download a structured copy of the data (ex. JSON, HTML), which you can then process and analyze with code. Other platforms do not, which means you may need to get creative – for example, manually recording ads or building your own dataset by hand. Be sure to give yourself enough time to research and identify a usable dataset. You may need to review several platforms before finding one that provides an adequate dataset.
- Keep in mind that some platforms take time to process data requests. This can range from a few minutes to 24 hours — or even up to a week. To avoid delays, **start your assignment early**. Some students have found that requesting a smaller data set (e.g., only data from the last 3 months or only advertising data) can result in faster turnaround times.

**Report output:** In your report, please document your results and findings:

1.1. Selected platform
1.2. Types of data collected by the platform
1.3. Selected dataset
1.4. Selected dataset's original size (count)
1.5. Selected dataset's reduced size (could be the same as above if reduction was not necessary)
1.6. The categories identified, examples of data that fall into that category
1.7. The relevancy buckets identified

**Step 2: Analyzing data**

Now that you have identified your dataset, the next step is to categorize and analyze it to gain a better understanding of its metrics. This involves:

- Classify your reduced dataset into the identified categories and relevancy buckets.
- Create a data flow graph using the Sankeymatic site (http://sankeymatic.com/build/) to create a visualization that connects your categories to the three relevancy buckets you've created.
- Compute basic statistical measures for each data category:
  - Total Count
  - Accuracy = Percentage of items in most relevant bucket compared to total count
  - Rubbish = Percentage of items in least relevant bucket compared to total count
- Identify which category was the most accurate (highest accuracy percentage) and which category was the most rubbish (highest rubbish percentage).
- Identify which data items fall under a regulated domain in law (as defined in the lectures). For each regulated domain, list in a table how many falls within that domain and provide a sample of

the associated data items. If no data item falls under a regulation domain, indicate "0" and explain how you reached that conclusion.

**Report output:** In your report, please document your results:
2.1. Data flow graph + Sankeymatic code
2.2. Statistical measures tables
2.3. Most accurate category, most rubbish category
2.4. Regulated domain table

**Step 3: Reflection**
Review the results of your report, including the research conducted, the dataset you identified, and the data analysis you performed (e.g., dataflow diagram, and the summary statistics table).

**Report output:** Reflect on your results and answer the following questions:
3.1. Were you surprised or not surprised by the amount of information collected by the platform? Why or why not? Consider any privacy implications.
3.2. Examine and discuss the accuracy or inaccuracy of the data collected. Was this the level of accuracy/inaccuracy you expected? Why or why not?
3.3. Will this information prompt any changes with how you interact with the platform? Why or why not?

**Step 4: Submission**
Submit a PDF report documenting your outputs for each question.

- **The report must follow JDF formatting.**
    - You can find a link to the JDF template here: JDF Templates. The Microsoft Word template is recommended to ensure proper formatting and styling.
    - Reports that do not follow JDF formatting, or that are not neat and well organized, may receive up to a 10% deduction, regardless of content quality.
- **Name your file exactly as follows: GTuserName_Assignment_1.pdf**
    - *Example: gBurdell3_Assignment_1.pdf*
    - Deductions may be applied if the file name is incorrect or does not follow this convention.
- **All charts, graphs, and tables must be generated using Python, Excel, or another suitable non-AI software application.**
    - Points may be deducted for charts or tables generated using AI tools.
- If you used a Jupyter Notebook (.ipynb) for your work, you may submit it as a supplementary file. However, note that **PDF report is required** and will be used for grading.

For your reference, an example report in JDF format is provided below. **Please note that this is an example report and it is incomplete. Please make sure you fully follow the instructions provided above.**

Sample: Data Profile Assignment

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# 1 PLATFORM RESEARCH AND DATA IDENTIFICATION

## 1.1 Selected Platform

Facebook

## 1.2 Types of Data Collect by the Platform

- Activity including ads, posts, and videos I've viewed or clicked "like" on
- Login activity including IP address, device used, the date and time
- List of advertisers currently using my information

## 1.3 Selected Dataset

List of advertisers currently using my information

## 1.4 Selected Dataset's Original Size

1700

## 1.5 Selected Dataset's Reduced Size

340

## 1.6 Categories

- Number of categories identified: 5
- Car Companies (e.g. International Autos Mercedes Benz)
- Social Impact (e.g. The National Association for the Education of Young Children)
- Shopping (e.g. Tiffany & Co.)
- Interest Groups (e.g. AARP)
- Entertainment – (e.g. Applebee's Grill & Bar)

## 1.7 Relevancy Data Buckets

- Yes
- No
- U Got To Be Kidding

## 2 DATA ANALYSIS

### 2.1 Dataflow Sankeymatic.com Script

```
FB Advertisers [136] Car Companies
FB Advertisers [68] Social Impact
FB Advertisers [34] Shopping
FB Advertisers [68] Interest Groups
FB Advertisers [34] Entertainment
Car Companies [17] Yes
Car Companies [119] No
Social Impact [68] Yes
Shopping [17] No
Shopping [17] U Got to be Kidding
Interest Groups [34] Yes
Interest Groups [34] No
Entertainment [17] Yes
Entertainment [17] No
```

### Dataflow Diagram

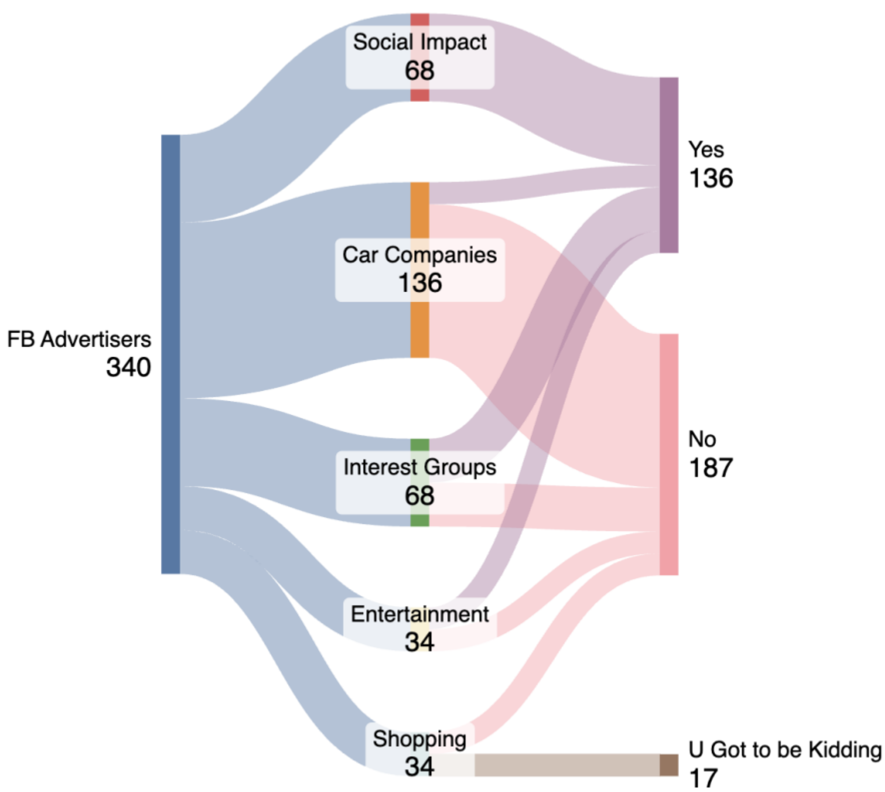


*Figure 1* — Dataflow diagram

### 2.2 Table: Summary Statistics (Note: Partial Example for One Category – Please complete this portion for *ALL categories*)

| Category | Data Bucket | Count | Accuracy | Rubbish |
|---|---|---|---|---|
| Shopping | U Got to be Kidding | 86 | | |
| | No | 84 | | |
| | Yes | 0 | | |
| | Total Count | 170 | 0% | 51% |

### 2.3 Most Accurate and Most Rubbish Category

My most accurate category: Social Impact

My most rubbish category: Shopping

### 2.4 Table: Regulated Domain Information

| Regulated Domain | Number of Items | Advertiser Sample |
|---|---|---|
| Credit | 230 | Alliant Credit Union<br>Anchor Capital |
| Education | 100 | Baylor College of Medicine<br>Daniels College of Business<br>Georgia State University |
| Employment | 0 | There were no advertisers related to employment, jobs or career opportunities. |
| Housing & Public Accommodation | 2 | Ashton Woods Homes<br>Echo Fine Properties |

## 3 REFLECTION

### 3.1 Reflection – Amount of data (Note: Partial example, please answer on your own)

I was very surprised by the amount of data Facebook had collected, especially because…

### 3.2 Reflection – Accuracy of data (Note: Partial example, please answer on your own)

My Facebook data had mixed accuracy results. For example…

### 3.3 Reflection – Changes made (Note: Partial example, please answer on your own)

Moving forward, I intend to…

## 4 REFERENCES